Sharp Covering-Index Decay for Schatten Classes

Automatic Math Discovery Run `fa_banach_001`

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Status

This packet records a **full result for the Schatten-class covering-index decay problem** left open in Kania–Maślany, *Raja's covering index of L_p spaces*, arXiv:2512.13249. It also records the optimal AUS power exponent of S_p. The result is limited to the classical Schatten classes $S_p(H) = L_p(B(H), \mathrm{Tr})$, not arbitrary non-commutative $L_p(M, \tau)$ spaces.

Source Passage

The source paper notes in Remark 5.3 that for $M = B(H)$ the non-commutative $L_p(M, \tau)$ space is the Schatten class S_p, records the lower bound $\Theta_{S_p}(n) \gtrsim n^{-1/r}$, $r = \min\{p, 2\}$, and then says that upper bounds are not pursued:

> **Proposition 5.2.** *Let (M, τ) be a semifinite von Neumann algebra such that $L_p(M, \tau)$ is infinite-dimensional, let $1 < p < \infty$, and set $r = \min\{p, 2\}$. Then there exists $c_p(M) > 0$ such that*
>
> $$\Theta_{L_p(M,\tau)}(n) \;\geqslant\; c_p(M)\, n^{-1/r} \qquad (n \geqslant 1).$$
>
> Thus the covering index of any non-commutative L_p-space decays at least like $n^{-1/r}$ with $r = \min\{p, 2\}$.
>
> *Remark* 5.3. If $M = L_\infty(\Omega, \mu)$ and $\tau(f) = \int_\Omega f\, d\mu$, then $L_p(M, \tau)$ is just $L_p(\mu)$ and Proposition 5.2 yields a lower bound of order $n^{-1/r}$ with $r = \min\{p, 2\}$. Our explicit covering in the commutative case (Proposition 3.1) shows that the optimal exponent is at most $1/p$; when $L_p(\mu)$ moreover admits an equivalent p-AUS norm, Corollary 3.3 gives $\Theta_{L_p(\mu)}(n) \asymp n^{-1/p}$. Thus the abstract non-commutative argument is not sharp even in the commutative setting when $p > 2$.
>
> If $M = B(H)$ with the canonical trace, then $L_p(M, \tau)$ is the Schatten class S_p on H. Proposition 5.2 then shows that
>
> $$\Theta_{S_p}(n) \;\gtrsim\; n^{-1/r}, \qquad r = \min\{p, 2\}.$$
>
> We do not pursue corresponding upper bounds in the non-commutative setting here.

The appendix then states the remaining sharp-decay problem explicitly:

$$\rho_{L_p(M,\tau)}(t) \le C_p t^r, \quad r = \min\{p,2\},$$

and hence $L_p(M,\tau)$ is asymptotically uniformly smooth of power type $r > 1$. In particular, the natural norm on $L_p(M,\tau)$ is r-AUS with $r = \min\{p, 2\}$. This is the input used in the proof of Proposition 5.2: once $L_p(M,\tau)$ is r-AUS, Raja's general theorem (Theorem 1.6) yields a power-type lower bound

$$\Theta_{L_p(M,\tau)}(n) \gtrsim n^{-1/r}.$$

Determining the *precise* optimal AUS exponent and the corresponding sharp covering-index decay rate for specific non-commutative L_p-spaces remains an interesting open problem, even for classical examples such as the Schatten classes $S_p = L_p(B(H), \mathrm{Tr})$.

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Definitions

For a Banach space X, the essential inradius of a set $A \subseteq X$ is

$$\varrho(A) = \sup\{r > 0 : \exists x \in A,\ \exists Y \le X \text{ finite codimensional},\ x + r(B_X \cap Y) \subseteq A\}.$$

Raja's covering index is

$$\Theta_X(n) = \inf\left\{ \max_{1\le k\le n} \varrho(A_k) : B_X = \bigcup_{k=1}^{n} A_k,\ A_k \text{ closed, convex, bounded} \right\}.$$

Let H be an infinite-dimensional Hilbert space. The Schatten class $S_p(H)$, $1 \le p < \infty$, is the space of compact operators T on H whose singular-value sequence belongs to ℓ_p, with norm $\|T\|_p = (\mathrm{Tr}\,|T|^p)^{1/p}$.

Proof Intuition

There are two complementary mechanisms. The upper bound is a matrix version of the scalar L_p block cover from the source paper. Split

$$H = H_1 \oplus \cdots \oplus H_n$$

and cover the unit ball according to which diagonal corner $P_k T P_k$ has small S_p-norm. Pinching onto diagonal corners is contractive, so one corner must have norm at most $n^{-1/p}$. The finite-codimensional inradius test is defeated by placing a rank-one operator far out in that corner, chosen to annihilate the finitely many functionals defining the subspace.

The lower bound is shorter. The diagonal operators form a 1-complemented copy of ℓ_p inside $S_p(H)$. The source paper records that covering index is monotone for contractively complemented subspaces, and it also gives the sharp scalar estimate $\Theta_{\ell_p}(n) \asymp n^{-1/p}$ for $1 < p < \infty$. Hence $S_p(H)$ cannot have faster covering-index decay than the diagonal ℓ_p part.

Sharp Covering-Index Decay

Theorem 1. *Let H be an infinite-dimensional Hilbert space and let $1 \le p < \infty$. Then for every $n \in \mathbb{N}$,*

$$\Theta_{S_p(H)}(n) \le n^{-1/p}.$$

Indeed, $B_{S_p(H)}$ has a cover by n closed convex sets, each of essential inradius exactly $n^{-1/p}$.

Proof. Fix $n \in \mathbb{N}$, and write

$$H = H_1 \oplus \cdots \oplus H_n$$

as an orthogonal direct sum of infinite-dimensional closed subspaces. Let P_k be the orthogonal projection onto H_k, and define

$$C_k(T) = P_k T P_k, \qquad T \in S_p(H).$$

Put $a = n^{-1/p}$, and define

$$A_k = \{T \in B_{S_p(H)} : \|C_k(T)\|_p \le a\} \qquad (1 \le k \le n).$$

Each A_k is closed, convex, and bounded.

Let $E(T) = \sum_{k=1}^n P_k T P_k$ be the finite pinching. If $U_\varepsilon = \sum_{k=1}^n \varepsilon_k P_k$ for $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{-1,1\}^n$, then

$$E(T) = 2^{-n} \sum_{\varepsilon \in \{-1,1\}^n} U_\varepsilon T U_\varepsilon.$$

Since the Schatten norm is unitarily invariant and convex, $\|E(T)\|_p \le \|T\|_p$. Since $E(T)$ is block diagonal,

$$\|E(T)\|_p^p = \sum_{k=1}^n \|C_k(T)\|_p^p.$$

Thus if $\|T\|_p \le 1$, at least one $\|C_k(T)\|_p$ is at most $n^{-1/p}$, and $B_{S_p(H)} = \bigcup_k A_k$.

Also $aB_{S_p(H)} \subseteq A_k$, so $\varrho(A_k) \ge a$.

We prove the reverse inequality. Fix k. Suppose that $T \in A_k$, $Y \le S_p(H)$ is finite codimensional, and $r > a$. We shall find $W \in B_{S_p(H)} \cap Y$ such that $T + rW \notin A_k$. Set $V = C_k(T)$, so $\|V\|_p \le a$.

Choose functionals $\Phi_1, \dots, \Phi_m \in S_p(H)^*$ with

$$Y = \bigcap_{j=1}^m \ker \Phi_j.$$

By the standard trace-duality representation of Schatten functionals, and using $S_1(H)^* = B(H)$ when $p = 1$, there are operators $B_j \in B(H)$ such that for rank-one operators $\theta_{u,v}(\xi) = \langle \xi, v\rangle u$ one has

$$\Phi_j(\theta_{u,v}) = \langle B_j u, v\rangle.$$

Let $R \le P_k$ be a finite-rank orthogonal projection on H_k. Since $V \in S_p(H_k)$, finite-rank compressions approximate V in S_p-norm; hence R may be chosen so that

$$\big(\|RVR\|_p^p + r^p\big)^{1/p} - \|V - RVR\|_p > a. \tag{1}$$

Indeed, the left side tends to $(\|V\|_p^p + r^p)^{1/p} > r > a$.

The subspace $K = H_k \ominus R(H_k)$ is infinite-dimensional. Pick a unit vector $u \in K$, then pick a unit vector

$$v \in K \cap \operatorname{span}\{B_1 u, \dots, B_m u\}^\perp.$$

This is possible because only finitely many vectors are excluded. Put $W = \theta_{u,v}$. Then W is supported in the k-th corner, $\|W\|_p = 1$, and $\Phi_j(W) = 0$ for every j, so $W \in Y$.

Since RVR acts on $R(H_k)$ and W acts from K to K, the operator $RVR + rW$ is block diagonal. Hence

$$\|RVR + rW\|_p^p = \|RVR\|_p^p + r^p.$$

Using (1),

$$\|C_k(T + rW)\|_p = \|V + rW\|_p \geq \|RVR + rW\|_p - \|V - RVR\|_p > a.$$

Thus $T + rW \notin A_k$. Therefore no finite-codimensional ball of radius larger than a is contained in A_k, and $\varrho(A_k) \leq a$. Since $\varrho(A_k) \geq a$, the covering has pieces of essential inradius exactly a, proving the theorem. $\square$

Theorem 2. *Let H be an infinite-dimensional Hilbert space and let $1 < p < \infty$. Then*

$$\Theta_{S_p(H)}(n) \asymp n^{-1/p}.$$

Proof. The upper bound is Theorem 1. For the lower bound, choose an orthonormal sequence (e_j) in H, and let $D \subset S_p(H)$ be the diagonal subspace

$$D = \left\{ \sum_j a_j \, e_j \otimes e_j : (a_j) \in \ell_p \right\}.$$

Then D is isometric to ℓ_p. The diagonal conditional expectation

$$\mathcal{E}(T) = \sum_j \langle Te_j, e_j \rangle \, e_j \otimes e_j$$

is contractive on $S_p(H)$; equivalently, it is the norm-one conditional expectation onto the diagonal masa, and it follows by finite pinching and approximation. Thus D is a contractively complemented subspace of $S_p(H)$.

Kania–Maślany's monotonicity lemma for contractively complemented subspaces [1, Lemma 3.3] gives

$$\Theta_{\ell_p}(n) = \Theta_D(n) \leq \Theta_{S_p(H)}(n).$$

Their scalar L_p estimate, applied to the counting measure, gives $\Theta_{\ell_p}(n) \asymp n^{-1/p}$ for $1 < p < \infty$ [1, Corollary 3.4 and the preceding remark]. Therefore

$$\Theta_{S_p(H)}(n) \gtrsim n^{-1/p}.$$

Together with the upper bound, this proves the claimed sharp decay. $\square$

Remark 1. *The upper-bound theorem also holds for $p = 1$. The lower-bound argument reduces the $p = 1$ endpoint to the known scalar value of Θ_{ℓ_1}; the source paper's non-commutative lower-bound discussion, however, is stated for $1 < p < \infty$, so the main sharp-decay statement above is formulated in that same range.*

Optimal AUS Exponent for S_p

Proposition 1. *Let $1 < p < \infty$. The largest AUS power exponent obtainable by an equivalent norm on $S_p(H)$ is*

$$\min\{p, 2\}.$$

Proof. The source paper recalls from the non-commutative Clarkson inequalities that the usual Schatten norm is r-AUS for $r = \min\{p, 2\}$ [1, Appendix]. Hence the optimal exponent is at least r.

It remains to exclude any $q > r$. AUSability passes to subspaces. If $1 < p \leq 2$ and $S_p(H)$ admitted an equivalent q-AUS norm for some $q > p$, then the diagonal subspace $D \cong \ell_p$ would be q-AUSable. Raja's lower bound, quoted as Theorem 1.6 in the source paper, would imply

$$\Theta_{\ell_p}(n) \gtrsim n^{-1/q}.$$

This contradicts the scalar estimate $\Theta_{\ell_p}(n) \asymp n^{-1/p}$, because $q > p$.

If $p \geq 2$, fix a unit vector $e_0 \in H$. The row space

$$R = \{\theta_{e_0,u} : u \in H\} \subset S_p(H)$$

is isometric to the Hilbert space H, since each $\theta_{e_0,u}$ has the single singular value $\|u\|$. If $S_p(H)$ were q-AUSable for some $q > 2$, then H would be q-AUSable. Raja's lower bound would give $\Theta_H(n) \gtrsim n^{-1/q}$, contradicting Kania–Maślany's exact Hilbert computation $\Theta_H(n) = n^{-1/2}$ [1, Theorem 2.2].

Thus no exponent larger than $\min\{p, 2\}$ is possible. $\square$

What This Solves and What It Does Not

For classical Schatten classes, the source paper's open Schatten covering decay question is resolved: the sharp power is $1/p$, i.e.

$$\Theta_{S_p(H)}(n) \asymp n^{-1/p} \qquad (1 < p < \infty).$$

The packet also gives the precise optimal AUS power exponent, $\min\{p, 2\}$. In particular, for $p > 2$, the sharp covering-index decay is governed by the diagonal ℓ_p subspace rather than by the optimal AUS exponent 2.

The packet does not settle arbitrary non-commutative $L_p(M, \tau)$ spaces. The upper-bound proof uses the concrete decomposition of H into matrix corners, and the lower-bound proof uses the diagonal ℓ_p masa in $B(H)$.

Verification and Novelty Check

The proof is abstract and uses no numerical computation. The code `code/make_open_problem_crops.py` renders the two source-passage crops from `source_paper.pdf`.

Local registry, solution, attempt, and proof-gap indexes had only the earlier partial packet from this run for arXiv:2512.13249, which has now been promoted and rewritten here. A bounded web/arXiv search on June 16, 2026 for "Raja's covering index Schatten $n^{-1/p}$", "Theta S_p covering index", "Schatten Raja covering index ell_p", and "essential inradius S_p Theta" found Raja's original paper arXiv:2503.03721 and the source paper arXiv:2512.13249, but no later paper giving the diagonal lower-bound upgrade or the resulting sharp Schatten decay. The source paper itself says that non-commutative upper bounds are not pursued and that the sharp Schatten-class decay remains open.

Revision History

The first pass on this target produced only the upper bound $\Theta_{S_p(H)}(n) \leq n^{-1/p}$. That superseded partial packet is kept inside this packet at `history/partial_upper_bound/`. The present version adds the diagonal ℓ_p lower bound and the optimal AUS exponent argument, so the indexed durable result is this full packet.

Human Review Recommendation

Review as a likely valid full solution of the Schatten covering-index decay subproblem. The main checks are the finite-codimensional rank-one tail argument in Theorem 1, the use of the diagonal conditional expectation and complemented-subspace monotonicity in Theorem 2, and the subspace obstruction in Proposition 1.

References

[1] T. Kania and N. Maślany, *Raja's covering index of L_p spaces*, arXiv:2512.13249.

[2] M. Raja, *A covering index for Banach spaces*, arXiv:2503.03721.

Human Review: Sharp Covering-Index Decay for Schatten Classes

Original automatic proof: `solution_packet.pdf`
Checked date: 19 June 2026
Status: Manually checked, focusing on Theorem 1.
Verdict: The proof of Theorem 1 appears correct.

Human Comments

I have checked Theorem 1 of the packet, which appears to be the new technical addition in the solution. The theorem proves the upper estimate

$$\Theta_{S_p(H)}(n) \leq n^{-1/p}$$

for the classical Schatten classes by constructing an explicit cover of the unit ball by the sets

$$A_k = \{T \in B_{S_p(H)} : \|P_k T P_k\|_p \leq n^{-1/p}\}.$$

The argument looks correct.

The covering step is clear. The finite pinching

$$E(T) = \sum_{k=1}^{n} P_k T P_k$$

is written as an average of unitary conjugates of T. Since Schatten norms are unitarily invariant and convex, $\|E(T)\|_p \leq \|T\|_p$. As $E(T)$ is block diagonal, this gives

$$\sum_{k=1}^{n} \|P_k T P_k\|_p^p \leq 1$$

for $T \in B_{S_p(H)}$, so at least one diagonal block has norm at most $n^{-1/p}$. Hence the sets A_k cover the unit ball.

The computation of the essential inradius of each A_k also appears correct. The easy inequality $\varrho(A_k) \geq n^{-1/p}$ follows from $n^{-1/p} B_{S_p(H)} \subseteq A_k$. For the reverse inequality, the proof uses a finite-codimensional subspace

$$Y = \bigcap_{j=1}^{m} \ker \Phi_j$$

and represents the functionals Φ_j by trace-duality operators B_j. One then chooses a rank-one operator $W = \theta_{u,v}$ supported in the infinite-dimensional tail of the k-th block, with

$$\langle B_j u, v \rangle = 0 \qquad (j = 1, dots, m),$$

so that $W \in Y$. This is the key finite-codimensional tail argument.

The use of a finite-rank projection $R \leq P_k$ is sound. Since finite-rank compressions approximate $V = P_k T P_k$ in Schatten norm, one can choose R so that

$$\left(\|RVR\|_p^p + r^p\right)^{1/p} - \|V - RVR\|_p > n^{-1/p}$$

for every $r > n^{-1/p}$. Then RVR and the chosen rank-one perturbation rW live on orthogonal summands, so

$$\|RVR + rW\|_p^p = \|RVR\|_p^p + r^p.$$

The reverse triangle inequality then gives

$$\|P_k(T + rW)P_k\|_p > n^{-1/p},$$

and therefore $T + rW \notin A_k$. This proves $\varrho(A_k) \leq n^{-1/p}$, completing Theorem 1.

Minor Amendment

There is one small wording issue in the proof. The sentence saying that the left-hand side tends to

$$(\|V\|_p^p + r^p)^{1/p} > r > n^{-1/p}$$

should be slightly weakened. If $V = 0$, then

$$(\|V\|_p^p + r^p)^{1/p} = r,$$

not strictly larger than r. The correct statement is

$$(\|V\|_p^p + r^p)^{1/p} \geq r > n^{-1/p},$$

which is fully sufficient for the argument.

Relation to the Lower Bound

The matching lower bound in the later sharp-decay statement seems essentially to come from the work of Kania–Maślany/Raja. Namely, the diagonal copy of ℓ_p inside $S_p(H)$ is contractively complemented, and the monotonicity of the covering index under contractively complemented subspaces transfers the known scalar lower estimate to the Schatten class. Thus the genuinely new technical point in the packet appears to be Theorem 1, i.e. the explicit Schatten upper-bound cover and the verification of the exact essential inradius of its pieces.

Review Outcome

Approve this packet as an apparently correct solution of the classical Schatten sharp-decay problem, with the minor inequality wording correction above. The proof of Theorem 1 appears correct, and the lower-bound side is consistent with the existing Kania–Maślany/Raja machinery.